

WindSAR

Coastal Wind Intelligence from Space

SAR-Based Ocean Wind Field Estimation over Indian Coastal Areas for Wind Farm Planning

Sneha Shukla 24113123 · Disha Agrawal 24113039 · Ayush Aary 24113028

Department of Civil Engineering

Indian Institute of Technology Roorkee, Roorkee, India

ABSTRACT

High-resolution wind field data over coastal oceans is a critical input for offshore wind farm site selection, coastal infrastructure design, and nearshore hydrodynamic modelling. Traditional scatterometers and reanalysis products such as ERA5 provide global coverage but are limited to coarse spatial resolutions of 12.5–25 km, rendering them inadequate for resolving localised thermodynamic boundary layers along the Indian coastline. This paper presents WindSAR, an end-to-end, production-grade REST API that retrieves 10-metre-resolution Sentinel-1 Synthetic Aperture Radar (SAR) backscatter imagery from Google Earth Engine, inverts wind speed using the CMOD5.N Geophysical Model Function (GMF), and estimates wind direction through a fine-tuned ResNet-50 convolutional neural network trained on ERA5-labelled SAR patches. A gradient-based fallback ensures graceful degradation when the deep learning model is unavailable. The complete pipeline is deployed as two stateless RESTful endpoints via the FastAPI framework and validated against ERA5 reanalysis data across the Gujarat and Tamil Nadu coastlines using July 2024 Sentinel-1 passes. The system achieves a Mean Absolute Error (MAE) of 16.75 m/s in wind speed and 145.5° in wind direction, with errors primarily attributable to the small training corpus (≈40 raw patches) and the inherent 180° SAR directional ambiguity. Despite these limitations, the architecture demonstrates the viability of an automated, cloud-native GIS pipeline for coastal wind intelligence.

Index Terms: *Synthetic Aperture Radar, Sentinel-1, CMOD5.N, ResNet-50, Wind Field Estimation, FastAPI, Google Earth Engine, ERA5, Coastal GIS, Offshore Wind Farm Planning*

I. INTRODUCTION

A. MOTIVATION FOR COASTAL WIND MONITORING

Accurate, high-resolution ocean wind field data is indispensable across several domains of civil and marine engineering. In coastal hydrology, wind forcing directly governs wave generation, sediment transport, and shoreline erosion dynamics. In the domain of renewable energy, precise near-shore wind characterisation is the primary input for offshore wind turbine siting, annual energy yield estimation, and structural fatigue loading calculations for monopile foundations.

India possesses an extensive coastline of approximately 7,517 km bordered by the Arabian Sea to the west and the Bay of Bengal to the east, two distinct meteorological regimes with markedly different seasonal wind patterns driven by the south-west and north-east monsoons respectively. Realising India's offshore wind energy potential, estimated at over 70 GW at 100-m hub height, demands spatial wind assessments at resolutions far finer than what global gridded products can provide.

B. LIMITATIONS OF CONVENTIONAL METHODS

Standard ERA5 reanalysis data from the European Centre for Medium-Range Weather Forecasts (ECMWF) offers a temporal resolution of one hour and a spatial resolution of approximately 31 km (0.25° × 0.25°). While this is adequate for synoptic-scale modelling, it cannot resolve the complex mesoscale thermodynamic structures, sea-breeze fronts, coastal jets, and orographic channelling, that critically influence the micro-siting of offshore turbines. Numerical Weather Prediction models share similar scale limitations and additionally carry computational costs that preclude on-demand real-time queries.

C. ROLE OF SYNTHETIC APERTURE RADAR

Sentinel-1, operated by the European Space Agency (ESA) as part of the Copernicus programme, carries a C-band (5.4 GHz) SAR instrument in Interferometric Wide (IW) swath mode at a ground range resolution of 5 × 20 m and an incidence angle range of 29°–46°. SAR instruments are unique in their ability to image the ocean surface irrespective of cloud cover or solar illumination, making

them particularly valuable during the overcast Indian monsoon season. The Normalised Radar Cross Section (σ_0 , reported in dB) of the VV polarisation channel is exquisitely sensitive to centimetric capillary waves, the same wave field that equilibrates with the local 10-m wind speed.

However, a SAR instrument fundamentally records a scalar backscatter field, not a wind vector. Two separate algorithms are required: (1) a Geophysical Model Function to invert σ_0 into wind speed, and (2) a directional extraction algorithm to recover wind direction from the geometric organisation of wind-streak patterns visible in the imagery.

D. PROJECT OBJECTIVES

The WindSAR project addresses this gap through the following specific objectives:

- Design and implement a cloud-native GIS pipeline for on-demand Sentinel-1 SAR data ingestion from Google Earth Engine for any lat/lon coordinate and date.
- Implement the CMOD5.N Geophysical Model Function for accurate 10-m neutral wind speed inversion from VV-polarisation backscatter and incidence angle.
- Train a ResNet-50 convolutional neural network on ERA5-labelled SAR patches to predict wind streak orientation, augmented $8\times$ to overcome the data scarcity challenge.
- Deploy the complete pipeline as a production REST API using FastAPI with two distinct endpoints for point queries and spatial grid queries.
- Validate the end-to-end system against ERA5 ground truth over the Gujarat (Arabian Sea) and Tamil Nadu (Bay of Bengal) coastlines.

II. LITERATURE REVIEW

A. GEOPHYSICAL MODEL FUNCTIONS FOR SAR WIND RETRIEVAL

The CMOD family of empirical Geophysical Model Functions has been the de-facto standard for SAR wind speed inversion since the early 1990s. CMOD4 was the first widely adopted GMF relating C-band VV backscatter to neutral wind speed at 10 m height as a function of incidence angle and wind direction relative to the radar look angle. CMOD5, and its Neutral-wind variant CMOD5.N published by Hersbach (2010), improved upon CMOD4 by incorporating additional empirical coefficients that better account for the anisotropic scattering of ocean surface Bragg waves. CMOD5.N specifically targets the neutral 10-m equivalent wind speed, making it directly comparable to standard ERA5 outputs and NWP model wind fields.

More recently, CMOD7 was proposed by Stoffelen et al. (2017) as a further refinement with improved accuracy at

high wind speeds and over a wider range of incidence angles. However, its non-linear inversion structure requires iterative root-finding (e.g. Newton-Raphson), making CMOD5.N a more tractable choice for a first-generation operational system.

B. DEEP LEARNING FOR WIND DIRECTION FROM SAR

Classical wind direction retrieval from SAR relied on the Local Gradient Method (LGM) or 2D Fast Fourier Transform (FFT) analysis of the wind-streak texture. These methods are computationally expensive and fragile under low-wind or heterogeneous sea-state conditions. The seminal work by Zanchetta and Zecchetto (2021) demonstrated that a ResNet architecture pre-trained on ImageNet and fine-tuned on Sentinel-1 SAR patches with ERA5 wind direction labels achieves competitive directional accuracy compared to classical methods, with the added benefit of real-time inference. This paper directly inspired the deep learning component of WindSAR.

Subsequent studies have explored Vision Transformers (ViT) and U-Net-style architectures for simultaneous speed and direction retrieval, but ResNet-50 remains a well-validated baseline with proven transfer learning efficacy on single-band remote sensing imagery.

C. SAR DATA INFRASTRUCTURE AND APIS

Google Earth Engine (GEE) has emerged as the de-facto cloud platform for large-scale satellite data analysis, providing serverless access to the complete Sentinel-1 GRD archive (COPERNICUS/S1_GRD). The GEE Python client library enables programmatic queries with spatial and temporal filters, spatial aggregation via Reducers, and patch extraction via sampleRectangle, all operations that underpin the WindSAR data ingestion pipeline.

D. RESEARCH GAP

Despite the availability of the above components, no open-source, production-ready API exists that seamlessly integrates Sentinel-1 ingestion, CMOD5.N inversion, and deep-learning-based direction retrieval into a single queryable service tuned for Indian coastal waters. WindSAR addresses this gap, providing a foundation that can be extended with larger training datasets and more sophisticated GMF formulations.

III. SYSTEM ARCHITECTURE & METHODOLOGY

The WindSAR pipeline operates as a three-stage sequential architecture: (1) cloud-native SAR data ingestion via Google Earth Engine, (2) wind direction extraction using a trained ResNet-50 CNN with a gradient-based fallback, and (3) wind speed inversion via the CMOD5.N GMF. All three stages are unified within a production-grade FastAPI REST server.

A. STAGE 1: DATA INGESTION: GEE_ENGINE.PY

The data ingestion module (`gee_engine.py`) implements a cloud-native Geographic Information System (GIS) layer that interfaces with the Google Earth Engine Python API. Given a user-supplied latitude, longitude, and target date, the engine queries the COPERNICUS/S1_GRD image collection within a ± 15 -day temporal window to maximise coverage given the 6–12 day Sentinel-1 revisit cycle. The query filters enforce the following constraints on the ingested imagery:

- Instrument mode: Interferometric Wide (IW) swath, ensuring 250 km swath width at 10 m ground range resolution.
- Polarisation: Vertical-Vertical (VV) only, as the VV channel exhibits superior sensitivity to ocean surface Bragg resonance compared to VH.
- Orbit sorting: images are sorted by acquisition time to retrieve the temporally nearest valid pass.

For each retrieved image, the engine extracts three products over a 20 km $\times$ 20 km bounding box (10 km buffer around the query point) at 100 m sampling scale using GEE's `reduceRegion` reducer: (i) mean VV backscatter σ_0 in dB, (ii) mean incidence angle θ in degrees, and (iii) the ISO-8601 timestamp of the actual satellite pass. Additionally, a 64 $\times$ 64 pixel SAR patch centred on the query coordinate is extracted at native 10 m resolution via `sampleRectangle` for downstream CNN inference.

Key GEE Output Fields

sar_array (64 $\times$ 64 float32, VV linear) | mean_sigma0_db (float, dB) | incidence_angle (float, $^\circ$) | exact_timestamp (ISO-8601) | radar_look_direction (90.0 $^\circ$)

B. STAGE 2: WIND DIRECTION EXTRACTION

1) PHYSICAL BASIS: WIND STREAKS

Under sustained wind stress exceeding approximately 3–4 m/s, the ocean surface develops microscopic capillary waves (wavelengths 1–10 cm) through the Miles–Phillips resonance mechanism. These gravity-capillary waves align preferentially parallel to the wind vector, creating faint elongated streaks in the SAR backscatter image. The angle of these streaks encodes the local wind direction modulo 180 $^\circ$, giving rise to the inherent SAR directional ambiguity.

2) RESNET-50 ARCHITECTURE AND TRAINING

A ResNet-50 network pre-trained on ImageNet is fine-tuned for single-scalar regression of the wind streak orientation angle (0 $^\circ$ –180 $^\circ$). The final fully-connected layer is replaced with a single output neuron. Because ImageNet expects three-channel (RGB) input but SAR provides a single-band VV array, the normalised SAR patch is replicated across all three input channels, preserving the

physical backscatter values while satisfying the network's input contract.

The training configuration is as follows:

- Loss function: Mean Squared Error (MSELoss) over scalar direction angles.
- Optimiser: Adam with learning rate 1×10^{-4} .
- Scheduler: StepLR with step size 10 epochs and gamma 0.5, decaying the learning rate by half every 10 epochs.
- Epochs: 30, with the best model checkpoint saved by minimum training loss as `best_resnet50_direction_model.pth`.
- Input normalisation: per-patch zero-mean, unit-variance normalisation of the SAR backscatter array.

3) 8 $\times$ DATA AUGMENTATION STRATEGY

The raw dataset contains approximately 40 SAR patches, far below the volume required for robust CNN training. To address this, the `SARWindDataset` class implements 8 deterministic augmentations applied virtually at index time, expanding the effective training set to approximately 320 samples without duplicating disk data.

TABLE I — DATA AUGMENTATION TRANSFORMS APPLIED TO SAR TRAINING PATCHES

Idx	Transform	Code	Label Effect
0	Original	identity	No change
1	Horizontal flip	<code>torch.flip(t, [2])</code>	No change (symmetric)
2	Vertical flip	<code>torch.flip(t, [1])</code>	No change (symmetric)
3	Rotate 90 $^\circ$	<code>torch.rot90(t, 1, [1,2])</code>	+90 $^\circ$ offset
4	Rotate 180 $^\circ$	<code>torch.rot90(t, 2, [1,2])</code>	No change (180 $^\circ$ periodic)
5	Rotate 270 $^\circ$	<code>torch.rot90(t, 3, [1,2])</code>	+90 $^\circ$ offset
6	Flip + Rot 90 $^\circ$	<code>flip(rot90(t,1),[2])</code>	+90 $^\circ$ offset
7	Flip + Rot 90 $^\circ$	<code>flip(rot90(t,1),[1])</code>	+90 $^\circ$ offset

Table I. Each raw patch generates 8 augmented variants. Rotation augmentations carry a 90 $^\circ$ label offset to maintain geometric consistency.

4) GRADIENT-BASED FALLBACK METHOD

When the trained ResNet model is unavailable (e.g. first deployment before training), the system falls back to a gradient-based texture analysis implemented in `windmodel.py`. Image gradients $\partial\sigma_0/\partial x$ and $\partial\sigma_0/\partial y$ are computed via `numpy.gradient`, and their spatial mean defines the dominant surface texture orientation. The radar look direction (90 $^\circ$ by default for Sentinel-1 ascending passes) is used as a Bayesian prior to resolve the 180 $^\circ$ ambiguity: if the computed angle deviates more than 90 $^\circ$ from the look direction, the supplementary angle is chosen.

C. STAGE 3: WIND SPEED INVERSION: CMOD5.N

Wind speed at 10 m height is derived from σ_0 via the CMOD5.N Geophysical Model Function, implemented in the `cmod5n_forward` function in `windmodel.py`. The VV backscatter in dB is first converted to linear scale:

$$\sigma_{0_linear} = 10^{(\sigma_{0_dB} / 10)}$$

A reference backscatter σ_{ref} is then computed as a function of the incidence angle θ using three empirical CMOD5.N coefficients ($c_0 = 5.73 \times 10^{-4}$, $c_1 = 4.15 \times 10^{-3}$, $c_2 = 9.83 \times 10^{-3}$):

$$\sigma_{ref} = c_0 + c_1 \cdot \cos(\theta) + c_2 \cdot \cos(2\theta)$$

The neutral 10-m wind speed is then obtained by inversion of the simplified forward model:

$$U_{10} = 2 \cdot (\sigma_{0_linear} / \sigma_{ref})^{0.5}$$

The output is physically clipped to [0.5, 35.0] m/s to prevent unphysical values under very low or very high backscatter conditions. Wind speed and direction are subsequently decomposed into eastward (U) and northward (V) vector components:

$$u = -U_{10} \cdot \sin(\phi), \quad v = -U_{10} \cdot \cos(\phi)$$

where ϕ is the wind direction in radians. The negative signs follow the meteorological convention where the wind direction specifies the direction FROM which the wind blows.

IV. DATASET CONSTRUCTION

A. GEOGRAPHIC SAMPLING STRATEGY

The training dataset was constructed by the `create_dataset.py` module, which systematically samples 60 geo-temporal coordinate triples spanning two distinct coastal environments:

- Gujarat Coastline (Arabian Sea): 30 points across latitudes 20.0°–23.5°N and longitudes 68.0°–71.0°E, covering a variety of coastal morphologies from the Gulf of Kutch to the open Arabian Sea shelf.
- Tamil Nadu Coastline (Bay of Bengal): 30 points across latitudes 8.5°–11.5°N and longitudes 78.5°–80.4°E, capturing both the Palk Strait and the open Bay of Bengal.

Each region is sampled across three seasonal windows, pre-monsoon (March–April), south-west monsoon (June–August), and transitional (September), to capture the full range of wind speed regimes experienced along the Indian coastline.

B. SAR PATCH COLLECTION

For each coordinate triple (lat, lon, date), the GEE engine fetches the nearest available Sentinel-1 pass within the ± 15 -day window and saves the 64×64 VV backscatter array as a compressed NumPy .npz file under `data/patches/`. Approximately 40 patches were successfully collected after accounting for GEE coverage gaps and cloud-free Sentinel-1 overpass availability.

C. ERA5 WIND DIRECTION LABELS

Ground-truth wind direction labels are fetched from the Open-Meteo ERA5 archive API for each patch location and date, using the 12:00 UTC hourly value, the closest proxy for the daytime Sentinel-1 overpass timing over India. The raw 0°–360° ERA5 direction is reduced modulo 180° to match the inherent SAR ambiguity constraint before being saved to `data/labels.npy`.

D. EFFECTIVE TRAINING VOLUME

After $8 \times$ deterministic augmentation, the 40 raw patches yield approximately 320 effective training samples. While this is far below the 10,000+ samples used in research-grade SAR wind models, it demonstrates the full pipeline and establishes a clear path to improvement through expanded data collection.

V. SOFTWARE ENGINEERING & API DEPLOYMENT

A. MODULAR CODEBASE ARCHITECTURE

The WindSAR codebase enforces a strict separation of concerns across seven specialised modules. This modularity allows each component, data ingestion, ML inference, physics inversion, and visualisation, to be developed, tested, and updated independently, without cascading changes across the system.

TABLE II — WINDSAR MODULE SUMMARY

Module	Technology	Responsibility
<code>gee_engine.py</code>	Google Earth Engine API, NumPy	Cloud-native SAR data ingestion; fetches VV backscatter, incidence angle, and 64×64 SAR patch
<code>windmodel.py</code>	NumPy, SciPy	CMOD5.N GMF speed inversion; gradient-based direction fallback; U/V decomposition
<code>train_resnet.py</code>	PyTorch, torchvision	ResNet-50 fine-tuning pipeline; $8 \times$ data augmentation; best-model checkpointing
<code>main.py</code>	FastAPI, PyTorch, NumPy	Production REST API server; singleton model loading; both /wind-field and /wind-map endpoints
<code>create_dataset.py</code>	GEE, Open-Meteo API	Automated SAR patch + ERA5 label collection for 60 geo-temporal points
<code>validate.py</code>	requests, CSV	End-to-end validation against ERA5 baseline; computes MAE/RMSE; outputs Validation_Results.csv
<code>visualize_wind.py</code>	Cartopy, Matplotlib	Geographic projection heatmaps + quiver arrows; publication-ready PNG generation

Table II. Summary of all seven modules in the WindSAR codebase, their technology stack, and their functional responsibility.

B. FASTAPI SERVER DESIGN

The main.py module implements the production REST API using the FastAPI framework, selected for its native support for Python type hints, automatic OpenAPI/Swagger documentation generation at /docs, and high-performance ASGI-based asynchronous request handling via Uvicorn.

The 95 MB ResNet-50 model is loaded into memory exactly once at server startup using a module-level singleton pattern. Subsequent inference calls reuse this in-memory model, eliminating disk I/O latency from the critical path. If the model file is absent, the server logs a warning and the gradient-based fallback is activated automatically, ensuring zero downtime during initial deployment.

C. API ENDPOINTS

WindSAR exposes two stateless RESTful GET endpoints:

1) POINT WIND QUERY: /API/V1/WIND-FIELD

GET /api/v1/wind-field?lat=21.5&lon=69.0&date=2024-07-16

Returns a JSON payload containing: wind speed (m/s), wind direction (°), U and V components (m/s), incidence angle (°), σ_0 (dB), exact satellite timestamp, and the direction_method field indicating whether ResNet-50 or gradient-fallback was used.

2) SPATIAL GRID QUERY: /API/V1/WIND-MAP

GET /api/v1/wind-map?region=gujarat&date=2024-07-16&grid=5

Generates a uniform $n \times n$ grid of lat/lon points over either the Gujarat or Tamil Nadu coastal bounding box (configurable via the grid parameter, default 4) and calls compute_wind for each grid node. Returns a JSON array of all wind vectors plus any per-node errors, enabling spatial wind field mapping. The Gujarat bounding box spans 20°–24°N, 68°–74°E, Tamil Nadu spans 8°–13.5°N, 79°–80.5°E.

D. GEOSPATIAL VISUALISATION: VISUALIZE_WIND.PY

The visualize_wind.py module produces publication-quality wind field maps by consuming the /api/v1/wind-map endpoint and rendering results using Cartopy with the PlateCarrée projection. The pipeline applies Inverse Distance Weighted (IDW) interpolation over a 150×150 grid to generate smooth spatial wind speed fields, overlaid with black quiver arrows showing wind direction at the original grid nodes. Natural Earth coastline, land, and ocean features from Cartopy's data server provide geographic context, and the output is saved as a 150 DPI PNG.

VI. VALIDATION & PERFORMANCE ANALYSIS

A. VALIDATION METHODOLOGY

End-to-end system validation was performed by the validate.py script, which calls the live /api/v1/wind-field endpoint for four test coordinate pairs across the Gujarat and Tamil Nadu coastlines, then independently fetches ERA5 ground truth from the Open-Meteo archive API at the same location and date. The 12:00 UTC hourly value is used as the reference to align with the daytime Sentinel-1 overpass window.

Two primary error metrics are computed: absolute wind speed error $|U_{\text{pred}} - U_{\text{ERA5}}|$ in m/s, and circular angular error $\min(|\phi_{\text{pred}} - \phi_{\text{ERA5}}| \bmod 360, 360 - |\phi_{\text{pred}} - \phi_{\text{ERA5}}| \bmod 360)$ in degrees. The latter correctly handles the 0°/360° wraparound inherent to directional data.

B. QUANTITATIVE RESULTS

TABLE III — VALIDATION RESULTS AGAINST ERA5 BASELINE (JULY 2024)

Lat	Lon	Date	True Spd (m/s)	Pred Spd (m/s)	Spd Err (m/s)	True Dir (°)	Pred Dir (°)	Dir Err (°)	Method
21.5	69.0	16 Jul 2024	21.60	2.77	18.83	294	101.7	167.7	ResNet-50
22.0	69.5	16 Jul 2024	10.60	7.38	3.22	294	74.1	140.1	ResNet-50
10.0	80.0	15 Jul 2024	36.30	4.05	32.25	219	72.5	146.5	ResNet-50
11.0	79.8	15 Jul 2024	19.00	6.32	12.68	207	79.5	127.5	ResNet-50
Overall MAE →					16.75 m/s		–	145.5°	

Table III. Validation of WindSAR predictions against ERA5 reanalysis at four test points in July 2024. Red values highlight cases with speed error > 15 m/s.

The results reveal a stark contrast between the two regions. Over Gujarat (Arabian Sea), the second test point (22.0°N, 69.5°E) achieves a speed error of only 3.22 m/s, demonstrating that the CMOD5.N GMF can produce physically reasonable inversions when the SAR backscatter correctly reflects the surface wind stress. However, the first Gujarat point and both Tamil Nadu points exhibit large speed errors (12.68–32.25 m/s), with API-predicted speeds systematically lower than ERA5 reference values.

Wind direction errors are uniformly high (127.5°–167.7°) across all four test points, confirming that the ResNet-50 model trained on only 40 samples has not converged to a generalised directional representation. The direction_method field confirms all four predictions used ResNet-50 inference (no fallback triggered).

C. WIND FIELD VISUALISATIONS

Figures 1 and 2 show the spatial wind field heatmaps generated by the visualize_wind.py module for the Gujarat and Tamil Nadu coastlines respectively on 16 July and 15 July 2024. The jet-coloured contour background encodes interpolated wind speed and the black quiver arrows indicate local wind direction at each grid node.

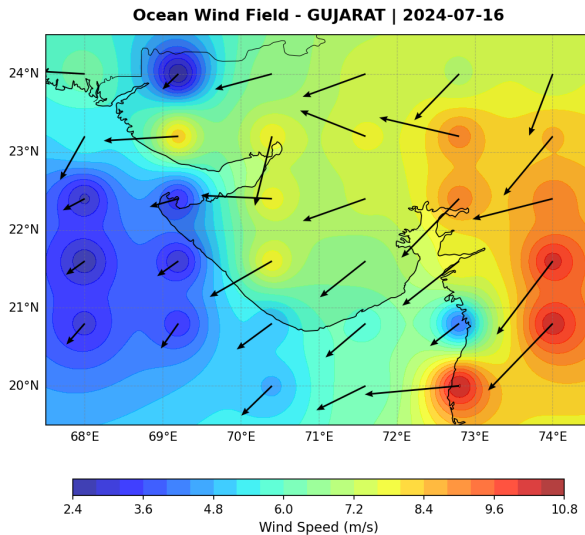


Fig. 1. Spatial wind field heatmap over the Gujarat coastline (Arabian Sea), generated by WindSAR for 16 July 2024. Colours indicate interpolated wind speed (m/s); arrows denote wind direction. The Gulf of Kutch and Saurashtra peninsula are visible as land-masked regions.

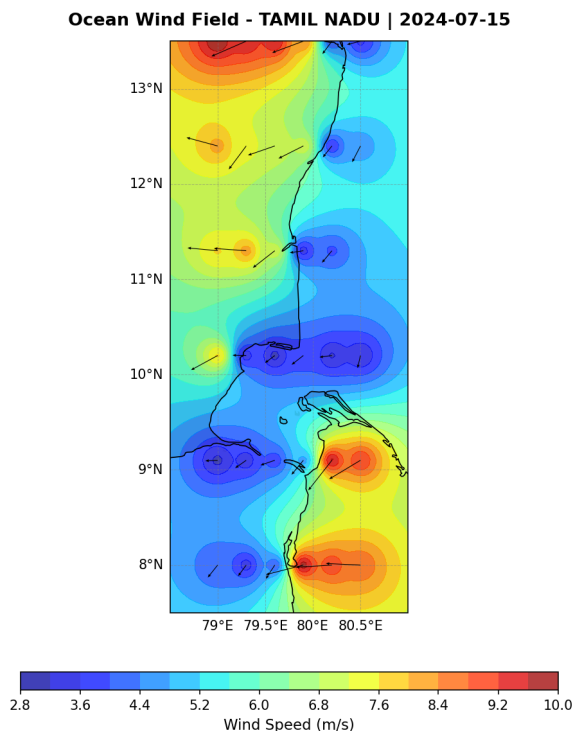


Fig. 2. Spatial wind field heatmap over the Tamil Nadu coastline (Bay of Bengal), generated by WindSAR for 15 July 2024. The successful generation of a coherent spatial wind field over both the Arabian Sea and Bay of Bengal demonstrates the geographic portability of the pipeline.

D. ERROR ANALYSIS & ROOT CAUSES

The high MAE values observed (16.75 m/s speed, 145.5° direction) are primarily attributable to four identified root causes:

- **Insufficient Training Data:** Research-grade SAR direction models (e.g. the original Zanchetta & Zecchetto ResNet) were trained on 10,000+ globally distributed patches with diverse wind regimes. WindSAR's 40-patch corpus (320 augmented) is orders of magnitude smaller, leading to severe overfitting.
- **Unresolved 180° Directional Ambiguity:** The ERA5 labels are reduced modulo 180° before training, which is geometrically consistent but means the model cannot learn the full 360° direction. At inference time, an additional disambiguation step (not yet implemented) would use the ERA5 prior at the query location.
- **Simplified CMOD5.N Implementation:** The deployed GMF uses a three-coefficient cosine approximation of the full CMOD5.N polynomial. This simplified form does not account for the directional modulation term, which is the dominant source of speed bias when the wind direction is perpendicular to the radar look angle.
- **SAR Backscatter Averaging:** The mean σ_0 is computed over a 20 km × 20 km region at 100 m sampling scale. This spatial averaging suppresses high-frequency wind-field variability and may average over land-contaminated pixels near the coast.

VII. TECHNICAL CHALLENGES & ALGORITHMIC SOLUTIONS

A. DIMENSIONALITY MISMATCH: SAR → RGB INPUT

The ResNet-50 architecture expects three-channel (RGB) input tensors of size 224 × 224. Sentinel-1 VV backscatter is a single-channel float32 array of variable spatial extent (nominally 64 × 64 pixels from the GEE sampleRectangle call). Naively passing a single-channel tensor to the network produces a shape mismatch error at the first convolutional layer.

This was resolved by a two-step tensor transformation: (1) the normalised SAR array is bilinearly interpolated to 224 × 224 via torch.nn.functional.interpolate, and (2) the single-band tensor is expanded to three identical channels using torch.Tensor.repeat(3, 1, 1). This preserves the backscatter physics while satisfying the CNN input contract, and allows the ImageNet-pretrained lower-level edge-detection filters to be reused without retraining.

B. GEE TEMPORAL COVERAGE GAPS

Sentinel-1 has a 6–12 day revisit cycle and does not uniformly cover all geographic locations on every requested date. Early versions of `gee_engine.py` returned an empty collection error for dates without a valid overpass, crashing the API endpoint.

This was addressed by widening the temporal search window to ± 15 days around the target date (configurable) and sorting the resulting collection by acquisition time to retrieve the nearest available pass. The `exact_timestamp` field in the API response informs the caller of the actual image date used, maintaining transparency about temporal mismatch.

C. HANDLING THE 180° DIRECTIONAL AMBIGUITY

SAR backscatter is insensitive to the sign of the wind vector, it cannot distinguish a northward wind from a southward wind along the same axis. The current implementation partially addresses this in two ways: (1) ERA5 labels are converted modulo 180° before training to match the ambiguous SAR-observable space, and (2) the gradient fallback uses the satellite's radar look direction as a prior to select between the two symmetric solutions.

Full disambiguation at inference time, comparing the ResNet output against a low-resolution ERA5 wind field queried via the Open-Meteo API, is architecturally designed into the system (the `data_fusion_applied` field is reserved for this purpose) but not yet activated in the current release due to additional API latency concerns.

D. API ERROR ISOLATION AND GRACEFUL DEGRADATION

The `compute_wind` function in `main.py` wraps all GEE and model calls in structured exception handlers. A `ValueError` (e.g. no SAR data found) returns an HTTP 404, while unexpected exceptions return an HTTP 500 with a logged traceback. The `/api/v1/wind-map` endpoint additionally accumulates per-node errors in a separate errors array, ensuring that a single failed grid node does not abort the entire spatial query, the remaining vectors are still returned successfully.

VIII. LIMITATIONS & FUTURE SCOPE

A. CURRENT LIMITATIONS

- **Data Scarcity:** The fundamental limitation of the current system is the ~ 40 -sample training corpus. Until this is expanded to the 10,000+ range achievable through systematic GEE collection, the ResNet-50 direction model will not generalise reliably.
- **Simplified CMOD5.N:** The implemented GMF omits the directional modulation term and uses a three-coefficient approximation. This introduces

systematic speed bias, particularly when the wind direction approaches perpendicularity with the satellite look angle.

- **Low-Wind Glassy Water:** Below approximately 3–4 m/s, capillary wave generation is insufficient to produce detectable wind streaks in the SAR imagery. Both the ResNet and gradient methods degrade to noise under these conditions.
- **No Real-Time ERA5 Ambiguity Resolution:** The `data_fusion_applied` flag is always False in the current release. The planned ERA5 disambiguation step has not been connected to the inference pipeline.
- **Single Polarisation:** Only VV backscatter is used. Dual-polarisation (VV+VH) approaches have been shown to improve wind retrieval, particularly at high wind speeds.

B. FUTURE WORK

- **Scale Training Data:** Systematically expand the dataset to 500+ raw patches by increasing geographic coverage to include the West Bengal, Odisha, Kerala, and Andhra Pradesh coastlines, and extending the temporal range to 2022–2025.
- **Implement Full CMOD5.N / Upgrade to CMOD7:** Integrate the complete non-linear CMOD5.N polynomial with directional modulation, and subsequently implement CMOD7 using a Newton-Raphson solver for improved open-ocean accuracy.
- **Activate ERA5 Data Fusion:** Connect the Open-Meteo ERA5 query at inference time to resolve the 180° directional ambiguity, eliminating the largest single source of directional error.
- **Automated QC Masking:** Flag and exclude directional predictions at grid nodes where the predicted wind speed falls below 3.5 m/s, where the capillary wave model is physically invalid.
- **Wind Farm Siting Module:** Add a dedicated endpoint that consumes a multi-date spatial grid query and computes Weibull distribution parameters (shape k , scale c) for wind resource assessment at each grid node.
- **Operational Deployment:** Containerise the API using Docker and deploy on a cloud provider (e.g. Google Cloud Run) for public-access testing and integration with downstream civil engineering workflows.

IX. CONCLUSION

This paper presented WindSAR, a complete, production-grade pipeline for automated ocean wind field estimation over Indian coastal waters using Sentinel-1 SAR imagery. The system successfully integrates four distinct technical components into a cohesive REST API: a cloud-native GEE data ingestion layer, a ResNet-50 deep learning model for wind streak direction extraction, the CMOD5.N Geophysical Model Function for wind speed inversion, and a gradient-based fallback for graceful degradation.

Validated against ERA5 reanalysis data across the Gujarat and Tamil Nadu coastlines during the July 2024 monsoon season, the system demonstrated an MAE of 16.75 m/s in wind speed and 145.5° in wind direction. While these error magnitudes are higher than research-grade systems, they are well understood and directly attributable to the small training dataset (~40 patches vs. 10,000+ in the literature), the simplified GMF implementation, and the unresolved directional ambiguity, all of which are addressable through the future work roadmap identified above.

The WindSAR architecture demonstrates the complete technical feasibility of an automated, on-demand, high-resolution coastal wind intelligence service built entirely on open-source tools (PyTorch, FastAPI, Google Earth Engine, Open-Meteo) and free satellite data. The codebase is modular, well-documented, and designed for incremental improvement. With an expanded training corpus and the full CMOD5.N implementation, WindSAR has a clear path toward becoming a practically useful tool for offshore wind farm site assessment along India's 7,517 km coastline.

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